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Flow injection Fourier transform infrared determination of caffeine in coffee

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Abstract

A fully automatized procedure has been developed for the Fourier transform infrared (FT-IR) spectroscopic determination of caffeine in coffee samples. The method involves the on-line extraction of caffeine with $CHCl_3$. Samples, weighed inside empty PTFE cartridges of 0.5 cm internal diameter (i.d.) and 1.5 ml volume, were humidified with four drops of 0.25 M NH_3. The cartridge was installed in a flow manifold, in which samples were extracted in a closed-flow system with 1 ml $CHCl_3$ during 6 min. Four hundred microliters of the extract were introduced in a microflow cell and absorbance measured as a function of time at 1659 cm^{-1}, with a baseline established between 1900 and 830 cm^{-1}, thus providing a fiagram. Peak height values of the FI recordings, obtained for samples, were interpolated in an external calibration line established from standard solutions of caffeine in $CHCl_3$. The method provided a limit of detection (LOD) of 9 mg l^{-1} caffeine, a relative standard deviation of 0.6% for five independent measurements of a solution containing 1 mg ml^{-1} and a sampling frequency of the whole procedure of 6 h^{-1}. Results obtained for market samples agree well with those found by the official chromatographic–spectrometric method, but involving a drastic reduction of solvents, from the 200 ml ether and 50 ml $CHCl_3$ required for each sample preparation by the reference procedure to less than 30 ml $CHCl_3$ necessaries for the whole determination of caffeine in a sample, also including standards and carrier solution.

Keywords: Caffeine determination; Coffee analysis; Flow injection analysis; Fourier transform infrared

1. Introduction

Caffeine is the main alkaloid present in coffee and thus the determination of caffeine is required in food laboratories in order to inform the consumers about the characteristics of coffee samples.

There are several methods proposed in the literature for the determination of caffeine in foods, based on ultraviolet (UV)–visible spectrophotometry [1–3], voltametry [4] or mass spectrometry [5]. Separation methods, like chromatography [6–8] or electrophoresis [9,10] have been also employed for the determination of caffeine in coffee, also being employed are

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gas chromatography and HPLC [6], ion chromatography [7], liquid chromatography–time of flight mass spectrometry [8], capillary electrophoresis [9] and micellar electrokinetic capillary electrophoresis [10].

The only precedents of the use of infrared (IR) spectrometry for caffeine determination in coffee are those proposed by Elisabeth et al. [11], who employed Fourier transform (FT) IR and RMN to confirm the identity of compounds separated by preparative supercritical fluid chromatography, and the paper of Fabian et al. [12] for FT-IR and NIR analysis of instant coffee mixtures, based on the use of PLS.

On the other hand, our preliminary results on liquid–liquid extraction of caffeine from coffee by $CHCl_3$ evidenced the possibility of doing FT-IR quantitative determinations of caffeine in extracted samples [13].

Automatic determination of analytes in natural samples is a challenge to today's analytical chemistry in order to improve the productivity of laboratories and to reduce, as much as possible, the reagent consumption and waste generation [14].

There are several automatic procedures proposed in the literature for the determination of caffeine in drugs [15,16], urine [17], soft drinks [18,19] and coffee [20–22] involving the determination of caffeine through UV–visible spectrometry [15,21], gas chromatography–mass spectrometry [17,20,21], amperometry [18] and FT-IR spectrometry [16,19].

However, the main part of published studies concerns the on-line treatment of liquid samples, or previously dissolved samples, being the fibre optic-based interface developed by Tena and Valcárcel [21], the single precedent for the on-line selective photometric determination of caffeine in solid samples, like ground coffee, after supercritical fluid extraction of the analyte.

The purpose of the present study is to develop a simple procedure for the fully automatic determination of caffeine in ground coffee through the on-line extraction of caffeine with $CHCl_3$. Extraction was carried out from the solid samples, weighed inside empty solid extraction cartridges and wetted with few drops of an aqueous solution of NH_3, and the on-line determination by FT-IR. Basically, the aforementioned procedure involves the consideration of coffee samples as a solid support in which caffeine was naturally concentrated, thus being making it possible to do the on-line elution of the analyte through its leaching with an appropriate solvent.

2. Experimental

2.1. Apparatus and reagents

A Mattson (Madison WI), model Research RSI FT-IR spectrometer, equipped with a DTGS detector and a KBr Ge coated beam splitter, with a water-cooled globar source, was employed to obtain the IR spectra with a nominal resolution of 8 cm^{-1} using a Spectra Tech (Warrington UK) microflow through cell with ZnSe windows and a pathlength of 0.457 mm.

The on-line caffeine extraction and FT-IR measurements were carried out using the manifold indicated in Fig. 1. Samples, weighed inside empty extraction cartridges of 0.5 cm internal diameter (i.d.) and 1.5 ml volume, were placed in a closed-flow system integrated by 0.3 m PTFE tubing of 0.8 mm i.d., and 20 cm length Viton (isoversinic) flexible tube of 3 mm o.d. and 1 mm i.d. also including a three-way directional valve for sampling, and a six-way Rehodyne type 50 (Cotati CA) PTFE injection valve with 1 ml loop. A Gilson Minipuls 2 peristaltic pump (Villier-le-bel, France) was used to assure the continuous extraction of caffeine. An additional peristaltic pump and another injection valve, equipped with a 400-μl loop, of the same type as those indicated, were employed to introduce discrete vol-

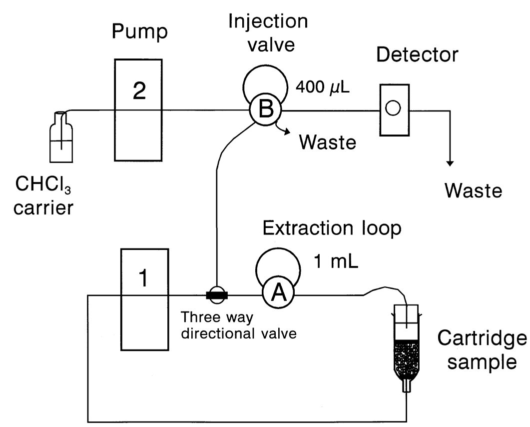

Fig. 1. Manifold employed for the FI-FT-IR determination of caffeine in coffee samples.

umes of the extracted samples into a carrier flow of $CHCl_3$ in order to be analyzed by FT-IR.

To do the study of the on-line caffeine extraction, the closed-flow system was modified by incorporating the flow through cell and deleting the carrier flow.

Caffeine was obtained from Fluka (Buchs, Switzerland). Chloroform, stabilized with 150 ppm amylene, was bought from Scharlau (Barcelona, Spain) and ammonium hydroxide 30% from Panreac (Barcelona, Spain). All reagents employed in this study were of analytical reagent grade. Chloroform, saturated with distilled water, was employed as a carrier and for the preparation of standards.

Roasted coffee samples and instant coffee, analyzed in this study, were obtained from the Spanish market.

2.2. Recommended procedure

The 0.1-g coffee sample was accurately weighed inside a PTFE cartridge of 0.5 cm i.d. and 1.5 ml volume. The sample was wetted with four drops of 0.25 M NH_3 aqueous solution, and then the cartridge was placed in the closed-flow manifold indicated in Fig. 1. Using the injection valve A, 1 ml $CHCl_3$ was introduced into the cartridge, from the bottom part, and after that the sense of the flow was reversed. The extraction of caffeine followed with a duration of 6 min using a flow rate of 1.18 ml min^{-1} and, after this time, 400 μl of the extract were sampled by using the three-way directional valve and the injection valve B. The extracted sample was injected into the carrier stream of $CHCl_3$ and transported through the measurement cell using a carrier flow of 1.2 ml min^{-1}.

Analytical measurements were carried out at 1659 cm^{-1}, corrected with a baseline established between 1900 and 830 cm^{-1}, and the corresponding FI recordings obtained as a function of time. The peak height of fiagrams of samples were interpolated in the external calibration line established from $CHCl_3$ solutions of caffeine injected through valve B and measured in the same conditions as the samples.

2.3. Reference procedure

One gram ground sample was accurately weighed, transferred to a 100 ml beaker and warmed on boiling water bath, for 2 min with 5 ml NH_4OH (4 M). After cooling, the sample was transferred to a 100-ml volumetric flask and diluted to the volume with H_2O, 5 ml of the turbid solution were mixed carefully with 6 g Celite 545, the mixture was placed inside a 25 × 250 mm tube column until a homogeneous and compact layer over a glass wool plug and a mixture of 3 g Celite 545 and 2 ml 2 M NaOH can be obtained. The aforementioned basic column was mounted above an acid column prepared from 3 ml 2 M H_2SO_4 and 3 g Celite 545, well mixed and placed between two glass wool layers inside a 25 × 250 mm tube. One hundred milliliters of water-saturated ether was sequentially passed through basic column to acid column and ether was discarded. Fifty milliliters of water-saturated ether was then passed through acid column and discarded. Forty-eight milliliters of water-saturated $CHCl_3$ was passed through acid column washing top of basic column with first portions and the caffeine solution recovered in a 50-ml volumetric flask, after which column elution was diluted to the mark with water-saturated $CHCl_3$ and the absorbance read at 276 nm against a water-saturated $CHCl_3$ blank, using caffeine $CHCl_3$ solutions as standards [23].

3. Results and discussion

3.1. FT-IR spectra of caffeine solutions and coffee extracts

Fig. 2 shows that both caffeine solutions in $CHCl_3$ and $CHCl_3$ extracts of coffee samples provide similar spectra in which, when other components are present in coffee extracts, it is possible to identify the main characteristic bands of caffeine at 1659 and 1554 cm^{-1}, in addition to the band at 1710 cm^{-1}, which unfortunately is interfered by other coffee components extracted with $CHCl_3$.

Through this study, the band at 1659 cm^{-1} was employed for controlling on-line the caffeine extraction, being that the most sensitive band of the analyte which can neither be interfered by the matrix, nor by the presence of water or NH_3 in the extracts.

3.2. Extraction of caffeine from coffee

Previous studies, carried out in batch, evidenced that to do a fast and complete extraction of caffeine

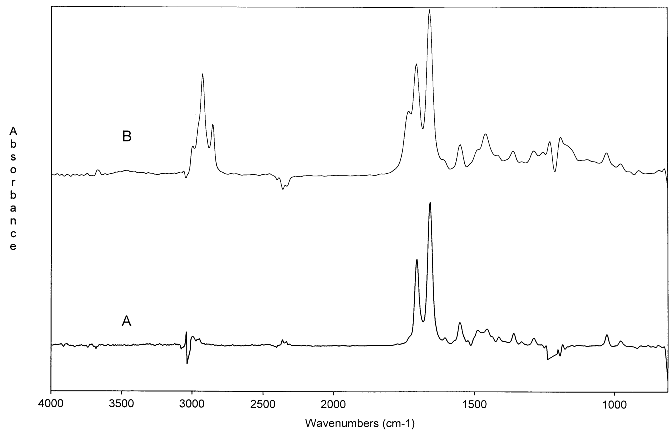

Fig. 2. FT-IR spectra of (A) 1 mg ml^{-1} caffeine solution in $CHCl_3$ and (B) a chloroformic extract of a roasted coffee sample containing $1.89 \pm 0.09\%$ caffeine. Both spectra were obtained for 16 cumulated scans at a resolution of 4 cm^{-1} and using the cell filled with water-saturated $CHCl_3$ as background.

from coffee samples, it is necessary to previously wet the samples with water, having observed that caffeine can be more easily leached from alkaline-wetted samples than from acid or neutral ones, thus

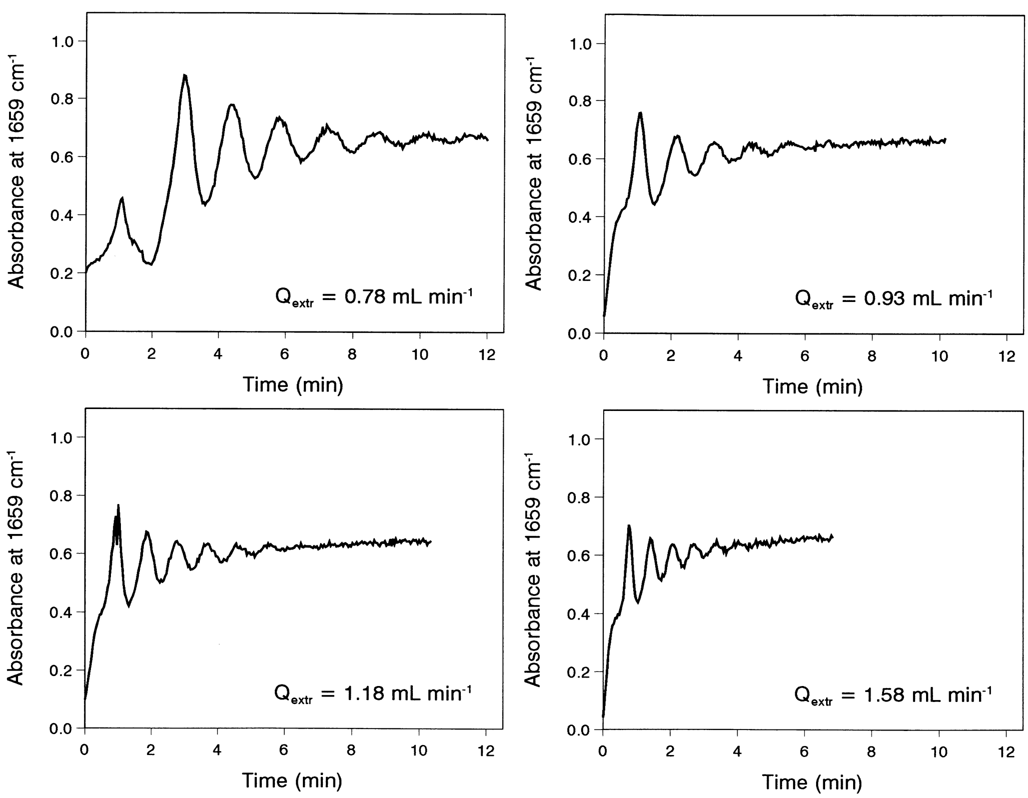

Fig. 3. Evolution, as a function of time, of the absorbance at 1659 cm^{-1} of 0.1 g roasted coffee sample extracted on-line with 1 ml $CHCl_3$ using a closed-flow system and different extraction flows.

Table 1
Effect of the extraction flow rate on the FI-FT-IR determination of caffeine in coffee. Experimental conditions: 0.1 g roasted coffee containing $1.89 \pm 0.09\%$ caffeine was extracted with 1 ml $CHCl_3$. Analytical data were obtained from absorbance measurements made at 1659 cm^{-1} corrected with a baseline established between 1900 and 830 cm^{-1} using two cumulated scans and a resolution of 8 cm^{-1}

Flow rate (ml min^{-1})	Caffeine (% w/w)	Stabilization time (min)
0.78	1.87 ± 0.03	11.4
0.93	1.83 ± 0.01	7.5
1.17	1.86 ± 0.02	6.3
1.58	1.91 ± 0.01	3.8

being selected a 0.25 M NH_3 aqueous solution for conditioning the samples previously to study their on-line extraction with $CHCl_3$.

In a preliminary study [13], we confirmed that a wetting time of 1 min with 2 ml NH_3 0.25 M, followed by a hand-shaking time of 5 min with 5 ml $CHCl_3$ and 10 min repose time, were good enough to obtain quantitative recoveries of caffeine from 0.5 g samples of roasted regular and instant coffee.

In the present study, three parameters, such as sample mass, $CHCl_3$ volume and extraction flow rate, were evaluated in order to find the faster and accurate conditions for caffeine on-line extraction.

The effect of the extraction flow rate was evaluated from 0.78 to 1.58 ml min^{-1}, using 0.1 g roasted coffee sample containing $1.89 \pm 0.09\%$ and 1 ml $CHCl_3$. The analytical data were obtained from two cumulated spectra at a resolution of 8 cm^{-1}.

Recordings on Fig. 3 correspond to the chemigrams obtained in the closed-flow system working at different extraction flows and measuring the absorbance at 1659 cm^{-1} and, as can be seen, upon increasing the extraction flow rate, the required time to obtain stable values of caffeine absorbance de-

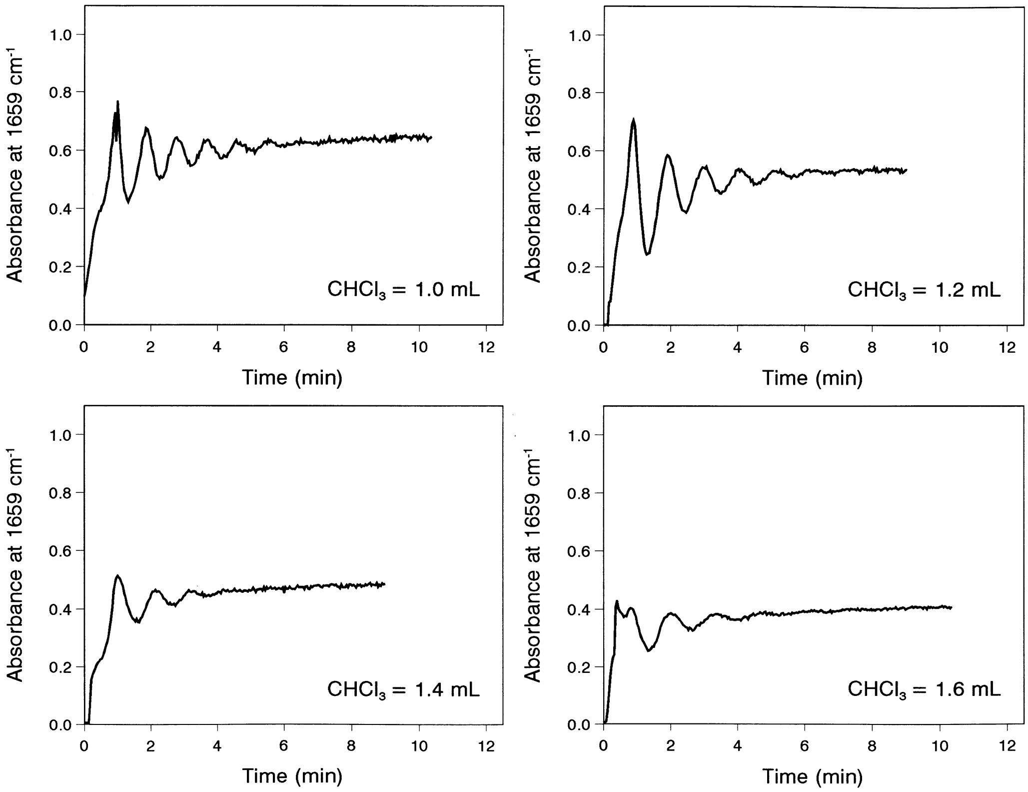

Fig. 4. Evolution, as a function of time, of the absorbance at 1659 cm^{-1} of 0.1 g roasted coffee extracts obtained on-line with different volumes of $CHCl_3$ using a closed-flow system and an extraction flow of 1.18 ml min^{-1}.

creases, (see Table 1) that being obvious in all cases where the on-line extraction of caffeine was complete. So, from these experiences, it can be concluded that a quantitative extraction of caffeine can be carried out only in 3.8 min by using an extraction flow of 1.58 ml min^{-1}. However, in order to obtain as rugged as possible methodology, a flow rate of the order of 1.18 ml min^{-1}, which involves an extraction time of approximately 6 min, was selected.

The effect of $CHCl_3$ was evaluated using a fixed mass of 0.1 g roasted coffee and an extraction flow of 1.18 ml min^{-1} and it was confirmed that on increasing the volume from 1 to 1.6 ml, the time required to obtain a total recovery of caffeine from coffee samples decreases from 6.3 to 4.1 min, as it can be seen in Fig. 4, which depicts the evolution, as a function of time, of absorbance values measured at 1659 cm^{-1} and in Table 2, which summarizes the data found.

In the aforementioned experiments, it must be noticed that the use of increasing volumes of $CHCl_3$, for a fixed coffee mass, reduces the maximum level of absorbance which corresponds to the quantitative extraction of caffeine. On the other hand, on comparing the absorbance–time curves depicted in Fig. 4 with those indicated in Fig. 3, for different extraction flows, it can be concluded that the increase of $CHCl_3$ volume improves the extraction of caffeine thus reducing the time required to do that and reducing also absorbance variations. However, in order to use as reduced as possible amount of chlorinated solvents, 1 ml $CHCl_3$ seems to be good enough for routine on-line analysis of coffee samples.

Table 2
Effect of the $CHCl_3$ volume on the FI-FT-IR determination of caffeine in coffee. Experimental conditions: 0.1 g roasted coffee containing 1.89 ± 0.09% caffeine was extracted at a flow rate of 1.18 ml min^{-1} using different volumes of $CHCl_3$. Absorbance measurements were carried out as indicated for Table 1

Chloroform volume (ml)	Caffeine (% w/w)	Stabilization time (min)
1.0	1.86 ± 0.02	6.3
1.2	1.85 ± 0.01	5.7
1.4	1.97 ± 0.01	4.4
1.6	1.86 ± 0.02	4.1

Table 3
Effect of the sample mass on the FI-FT-IR determination of caffeine in coffee. Experimental conditions: Different coffee mass values of a sample containing 1.89 ± 0.09% caffeine were extracted on-line using a closed-flow system with 1 ml $CHCl_3$ and an extraction flow of 1.18 ml min^{-1} during an extraction time of 6 min. Absorbance measurements were carried out in the same conditions than those reported for data in Table 1 and 2

Sample (mg)	Caffeine (% w/w)
53.1	1.94 ± 0.01
78.8	1.93 ± 0.02
101.6	1.86 ± 0.02
122	1.85 ± 0.03
151.7[a]	1.08 ± 0.64

[a] In this case, the need of using an increased volume of 0.25 M aqueous NH_3 for sample wetting created some troubles in the flow-through cell, due to the introduction of small drops of water which were not observed in other cases.

The effect of sample mass on the caffeine extraction with 1 ml $CHCl_3$ and using an extraction flow of 1.18 ml min^{-1} was evaluated from 53 to 152 mg coffee and, as can be seen in data of Table 3, the increase of the sample mass tends to reduce the quantitativity of caffeine determination, carried out by on-line extraction and FT-IR after 6 min extraction in the closed-flow system. So, through this study, a sample mass of 0.1 g was employed.

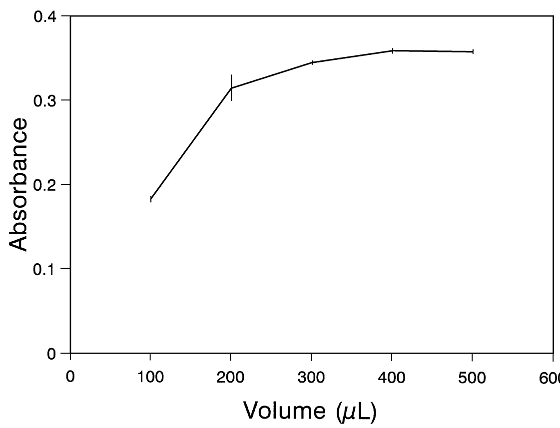

Fig. 5. Effect of injection volume on the absorbance at 1659 cm^{-1} obtained for extracts of a coffee sample. Experimental conditions: carrier flow: 0.86 ml min^{-1}, two cumulated scans and a resolution of 8 cm^{-1}.

Table 4
Analytical figures of merit of the FI-FT-IR determination of caffeine. In all the measurements the baseline was established between 1900 and 830 cm^{-1}. LOD: Limit of detection established for $K = 3$ (a probability level of 99.6%). RSD: Relative standard deviation for five independent measurements carried out at a concentration level of 1 mg ml^{-1}. The upper limit of the dynamic range was established from the adjustment of data in order to obtain a linear calibration

Wave number (cm^{-1})	$A = (a \pm s_a) + (b \pm s_b)C$ (C in mg ml^{-1})	r	LOD (mg l^{-1})	RSD (%)	Dynamic range (mg ml^{-1})
1659	$(0.05 \pm 0.01) + (0.302 \pm 0.007)C$	0.9992	9	0.6	0.25–2.5
1554	$(0.009 \pm 0.002) + (0.059 \pm 0.001)C$	0.9995	50	4.2	0.25–2.5

3.3. Effect of the FI parameters on FT-IR caffeine determination

After on-line extraction from solid samples, extracts were injected in a carrier flow of $CHCl_3$ and absorbance peak height was measured at 1659 cm^{-1}.

Two experimental parameters, such as the injected sample volume and the carrier flow, can drastically affect the analytical sensitivity obtained in FI conditions as compared with those found for the batch measurements.

In order not to sacrifice the analytical sensitivity, the effect of the sample injection volume on the absorbance values was evaluated and found at 1659 cm^{-1} and, as can be seen in Fig. 5, a volume of 400 μl of extracts permits to reach the steady state conditions for a carrier flow of 0.86 ml min^{-1} and, on the other hand, it was confirmed that for this sample volume injected, the analytical sensitivity does not vary for a flow rate ranging from 0.3 to 1.5 ml min^{-1}. Because of that, a carrier flow of 1.2 ml min^{-1} was selected. In these conditions, a sampling frequency of 103 h^{-1} was obtained, which means that no more than 20 ml $CHCl_3$ are required to obtain a calibration curve of five standard solutions, each one measured three times, thus providing a total consumption of $CHCl_3$ of less than 30 ml for both sample preparation and calibration.

3.4. Analytical figures of the FI determination of caffeine in coffee

Table 4 summarizes the main features of the methodology developed through this study for caffeine determination in coffee samples. As it can be seen using both 1659 and 1554 cm^{-1} bands, caffeine could be determined in coffee samples. However, it is clear that the use of the band at 1659 cm^{-1} provides the best sensitivity and repeatability, having obtained a LOD of 9 mg l^{-1}, which compares well with the 3 mg l^{-1} LOD obtained by off line extraction and stopped flow determination [13], and a variation coefficient of 0.6% for five independent measurements of a solution containing 1 mg ml^{-1} caffeine.

In the aforementioned conditions, which involve an extraction time of 6 min in the closed-flow sys-

Table 5
Recovery of caffeine from spiked commercial coffee samples. Note: Values indicated correspond to the average recovery ± the standard deviation of three independents determinations

Caffeine added (mg)	Caffeine found (mg)	Recovery percentage (%)
1.44	1.40 ± 0.13	97 ± 9
1.58	1.60 ± 0.11	101 ± 7
1.84	1.85 ± 0.10	101 ± 5
2.05	1.94 ± 0.11	95 ± 5

Table 6
Determination of caffeine in commercial coffee samples

Sample	Caffeine percentage (% w/w)		
	Official method [23]	FI-FT-IR	Batch FT-IR [13]
Regular coffee A	1.89 ± 0.09	1.85 ± 0.02	1.851 ± 0.007
Regular coffee B	1.78 ± 0.03	1.75 ± 0.05	1.74 ± 0.01
Regular coffee C	1.76 ± 0.08	2.00 ± 0.02	1.898 ± 0.002
Instant coffee	3.9 ± 0.1	3.91 ± 0.09	4.101 ± 0.008

tem and the use of a carrier flow of 1.2 ml min^{-1} and an extract injection volume of 400 μl, it is clear that the sampling frequency of the whole procedure is of the order of 6 h^{-1} including the use of less than 30 ml $CHCl_3$ for the determination of caffeine in a sample, also including standard and carrier solutions, thus providing a faster and environmentally friendly methodology than that employed as official reference method, which requires more than 2 h for sample preparation and 200 ml ether and 50 ml $CHCl_3$ for each sample.

Additional studies were carried out to test the accuracy of the FI-FT-IR procedure by evaluating the recovery of caffeine added to market samples of coffee. As it can be seen in Table 5, samples spiked with known amounts of caffeine, from 1.44 to 2.05 mg, were accurately analyzed by the aforementioned procedure, having obtained an average recovery percentage of 99%.

3.5. Analysis of commercial samples

The developed procedure for the on-line extraction and FI-FT-IR determination of caffeine in coffee was employed to analyze three regular roasted coffee samples of Arabica type South American coffee mixtures and one instant coffee, using caffeine solutions in $CHCl_3$ as standards. Results obtained are summarized in Table 6 and, as can be seen, caffeine concentrations found are comparable to those obtained by a reference official procedure based on chromatographic and spectrometric determination at 276 nm [23] and with those obtained in batch by FT-IR [13], thus indicating that the FI-FT-IR measurements provide an accurate methodology but with a clear reduction of the time required for sample preparation, reagent consumption and waste generation, as indicated before.

On the other hand, on comparing the developed methodology with the previous in batch analysis by off-line extraction and stopped flow FT-IR determination, the method developed through this study reduces the consumption of $CHCl_3$ from 5 to 1 ml for each sample extraction and the time of extraction from 16 to 6 min.

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